Power System Economics – exam questions for 2021

The Answers

Question 1

(a) Describe the significance of the concept of system marginal price and explain why it varies with time in the case of electricity.

[2]

(b) Three generators are available to supply demand of $D = 600\ [MW]$. The cost of generating power $C_i(P_i)$ corresponding to each of the generators is:

$$C_1(P_1) = 800 + 2 * P_1 + 0.05 * P_1^2\ [\frac{£}{h}]$$

$$C_2(P_2) = 650 + 8 * P_2 + 0.25 * P_2^2\ [\frac{£}{h}]$$

$$C_3(P_3) = 500 + 16 * P_3 + 0.55 * P_3^2\ [\frac{£}{h}]$$

Calculate the optimal production of each generator, the system marginal cost, the system average cost and the profits of each generator.

[6]

(c) How would the dispatch change if three generators have the opportunity to buy some of this energy on the spot market at a price of 42.60 £/MWh? How much energy will be bought from the spot market?

[3]

(d) Assume that the minimum and maximum output limit of each generator is:

$$P_1^{min} = 100\ [MW],\ P_1^{max} = 600\ [MW]$$

$$P_2^{min} = 60\ [MW],\ P_2^{max} = 120\ [MW]$$

$$P_3^{min} = 40\ [MW],\ P_3^{max} = 100\ [MW]$$

(i) Determine the optimal production of each generator, the system marginal cost, the system average cost and the profits of each generator.

[4]

(ii) Based on the results, briefly explain the impact of considering these maximum output limits on the total cost of operating the system and the profits of the three generators.

[2]

Answers

(e) In the solution of **(d)**, the profit of one of the generators is negative. Explain why this happens, what this means for this generator, as well as why and how it should be avoided in practice.

[3]

Question 2

(a) In the context of the competitive market concept, answer the following questions:

(i) What are the conditions for perfect competition?

[2]

(ii) How can a company in an imperfect competition environment influence the market price?

[1.5]

(iii) What is the relationship between the marginal cost of production and the market price in perfectly and imperfectly competitive markets?

[1.5]

(b) Consider a market for electrical energy that is supplied by two generating companies with the two corresponding cost functions:

$$C_A(P_A) = 26 * P_A + 0.5 * P_A^2 \ [\frac{£}{h}]$$

$$C_B(P_B) = 30 * P_B + 0.3 * P_B^2 \ [\frac{£}{h}]$$

The inverse demand curve for this market is estimated to be:

$$\pi = 200 - 1.5 * D \ [\frac{£}{MWh}]$$

Assuming perfect competition, calculate:

(i) electricity price and level of demand

[1.5]

(iii) production levels and the profits made by the generating companies.

[1.5]

(c) Assuming a Cournot model of competition:

(i) Form a table to calculate the Nash equilibrium point of this market, i.e. market price, demand quantity and profit of each company for different levels of productions for each of the companies. Consider range of productions of company A at 35MW, 37MW and 39MW and of company B at 37MW, 39MW and 41MW.

[8]

(ii) Based on the results, briefly explain the impacts of imperfect competition on the market outcome.

[2]

(d) Explain the principal difference between Bertrand and Cournot competition models.

[2]

Answers

Question 3

For the system shown in Figure 1, and the accompanied data given in Tables 1, 2 and 3, answer the following questions:

(a) Ignoring the impact of transmission constraints, calculate:

(i) minimum cost dispatch of the generators

[2]

(ii) the corresponding network flows and show that line 1-2 would be overloaded.

[2]

(b) This line overload can be eliminated by *A)* increasing generation at node 3 and reducing generation at node 1; or by *B)* increasing generation at node 2 and reducing generation at node 1.

(i) Calculate the hourly cost of options *A)* and *B)* above. Which of the two options is preferable and why?

[6]

(ii) Calculate the nodal prices when the more preferable of the two options compared in **(i)** is implemented.

[4]

(c) In order to manage the exposure to price fluctuations, Consumer (Load) connected to bus 2 has made a contract for difference with generation at bus 1 at a strike price of 26.5£/MWh and purchased a transmission congestion contract that gives the right to transport 90MW from bus 1 to bus 2. Calculate:

(i) income from the transmission congestion contract received by the Consumer (Load) at bus 2

[3]

(ii) net cost and corresponding unit price that demand at bus 2 will pay for the energy consumed under this condition.

[3]

Answers

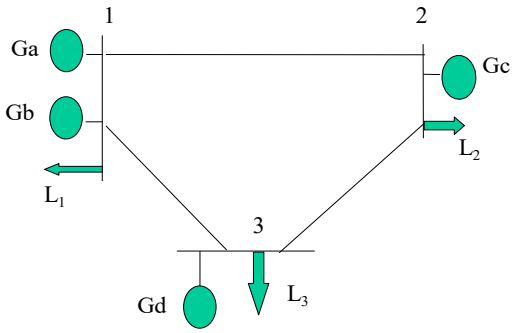


Figure 1: System structure

Table 1 Load data

Node	Load (MW)
1	80
2	90
3	330

Table 2 Generator data

Generator	Capacity (MW)	Marginal cost (£/MWh)
Ga	170	19
Gb	345	15
Gc	110	35
Gd	105	25

Table 3 Line data

Line	Per unit reactance	Capacity (MW)
1-2	0.3	160
1-3	0.3	260
2-3	0.2	110

Answers

Question 4

(a) Two areas, A and B, of the Borduria power system are connected by a 1200MW transmission link operated by the Borduria Transco. System load is concentrated in area B. In winter, the load is 3,500MW while summer load is 2,000 MW. The cost of generation in the areas can be modelled by the following expressions:

$C(P_A) = 1 \cdot P_A + 0.001 \cdot P_A^2$ [£/h] for area A

$C(P_B) = 2 \cdot P_B + 0.002 \cdot P_B^2$ [£/h] for area B

Determine the optimal levels of generation in areas A and B for each of the seasons neglecting the impact of the transmission system.

[4]

(b) If necessary, modify the levels of generation computed in (a) to take into consideration the capacity of the transmission link. For each of seasons:

(i) What are the electricity prices in areas A and B?

[2]

(ii) Calculate the generation costs in areas A and B.

[2]

(c) Assuming that the duration of the winter and summer period is 2,700h and 6,060h respectively, calculate:

(i) The total annual generation costs

[2]

(ii) The total annual revenue received by the generators and the charge imposed on demand if electricity is priced at locational short-run marginal prices.

[2]

(iii) The total annual transmission revenue received by Borduria Transco in this case.

[2]

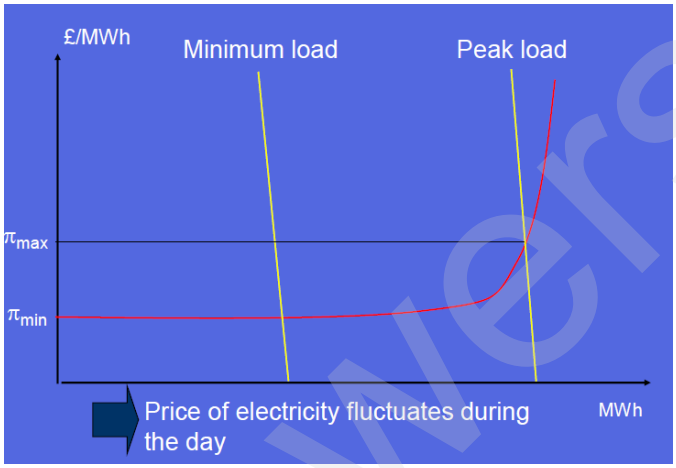
(d) Borduria Transco has proposed to the Borduria Regulator to reinforce the line between the two areas by 900 MW. Assuming that the annuitised investment cost of the additional line is 5,000,000 £/year, determine if this proposed investment is justified.

[6]

Answers

Solution to Question 1

(a) The system marginal price signifies the marginal cost of serving an additional unit of demand at the least-cost dispatch of the total load to the available generators. In contrast with other commodities, electrical energy cannot be economically stored in large quantities and thus it must be produced exactly when it is consumed by the inflexible demand. Moreover, the electrical demand is not the same every hour, day, month etc and the generation side is characterized by a production cost which varies with the size of the load. The combination of these effects explains why the system marginal price varies with time. This is demonstrated in the figure below. The supply curve of the generation side exhibits a "hockey-stick" shape, being rather flat at the area of low loads, where cheap base generators are employed for the satisfaction of the demand, and rather steep at the area of high loads, where expensive peaking generators are also employed to cover the demand. In combination with the fluctuation of the demand from a minimum to a peak value, temporal variations in the system marginal price emerge.



[2 marks; 1 mark for describing the significance of system marginal price, 1 mark for explaining why marginal price varies with time in the case of electricity]

(b) In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost λ yielding:

$$\frac{\partial C_1(P_1)}{\partial P_1} = 2 + 0.1 * P_1 = \lambda \rightarrow P_1 = \frac{\lambda - 2}{0.1} \quad (1.1)$$

$$\frac{\partial C_2(P_2)}{\partial P_2} = 8 + 0.5 * P_2 = \lambda \rightarrow P_2 = \frac{\lambda - 8}{0.5} \quad (1.2)$$

$$\frac{\partial C_3(P_3)}{\partial P_3} = 16 + 1.1 * P_3 = \lambda \rightarrow P_3 = \frac{\lambda - 16}{1.1} \quad (1.3)$$

The satisfaction of the system demand implies that:

$$P_1 + P_2 + P_3 = D = 600\ [MW] \quad (1.4)$$

Substituting (1)-(3) in (4) gives for the system marginal cost:

$$\frac{\lambda - 2}{0.1} + \frac{\lambda - 8}{0.5} + \frac{\lambda - 16}{1.1} = 600 \rightarrow \lambda = 50.39\ [\frac{£}{MWh}] \quad (1.5)$$

Substituting (5) in (1)-(3) yields for the optimal production of each generator:

$$P_1 = 483.94\ [MW] \quad (1.6)$$

$$P_2 = 84.79\ [MW] \quad (1.7)$$

$$P_3 = 31.27\ [MW] \tag{1.8}$$

Substituting (6)-(8) in the cost functions of the generators yields for the total cost of operating the system:

$$C_{tot} = C_1(P_1) + C_2(P_2) + C_3(P_3) = 18141.55\ [£/h] \tag{1.9}$$

The system average cost is:

$$\frac{C_{tot}}{P_1+P_2+P_3} = 30.24\ [\frac{£}{MWh}] \tag{1.10}$$

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega_1 = \lambda * P_1 - C_1(P_1) = 10910.07\ [£/h] \tag{1.11}$$

$$\Omega_2 = \lambda * P_2 - C_2(P_2) = 1147.28\ [£/h] \tag{1.12}$$

$$\Omega_3 = \lambda * P_3 - C_3(P_3) = 37.71\ [£/h] \tag{1.13}$$

[6 marks; subtract 2 marks if the calculations of optimal production are wrong; subtract 1 mark if the system marginal cost is wrong; subtract 1 mark if the system average cost is wrong; and subtract 2 marks if the calculation of generators' profits are wrong]

(c) It would produce up to the point where the marginal cost of each unit is 42.60 £/MWh. The rest will be bought from the market at 42.60 £/MWh.

$$2 + 0.1 * P_1 = 42.60 \rightarrow P_1 = \frac{42.60-2}{0.1} = 406.00\ [MW] \tag{1.14}$$

$$8 + 0.5 * P_2 = 42.60 \rightarrow P_2 = \frac{42.60-8}{0.5} = 69.20\ [MW] \tag{1.15}$$

$$16 + 1.1 * P_3 = 42.60 \rightarrow P_3 = \frac{42.60-16}{1.1} = 24.18\ [MW] \tag{1.16}$$

So the rest 600 – 406.00 – 69.20 – 24.18 = 100.62 MW will be bought from the spot market.

[3 marks; subtract 2 marks if the calculations of optimal dispatches are wrong; subtract 1 mark if the energy bought from the spot market is wrong]

(d-i) Given that $P_3 < P_3^{min}$ the solution calculated in **(b)** is not feasible when taking into account given the minimum output limits of the generators. The optimal solution can now be calculated by fixing the output of generator 3 to:

$$P_3' = P_3^{min} = 40\ [MW] \tag{1.17}$$

and satisfying the remaining demand:

$$D' = D - P_3' = 560\ [MW] \tag{1.18}$$

through the remaining generators 1 and 2. Following the same method as in (b), we get:

$$\frac{\lambda'-2}{0.1} + \frac{\lambda'-8}{0.5} = 560 \rightarrow \lambda' = 49.67\ [\frac{£}{MWh}] \tag{1.19}$$

Substituting (19) in (1)-(2) yields for the optimal production of each generator:

$$P_1' = 476.67[MW] \tag{1.20}$$

$$P_2' = 83.33\ [MW] \tag{1.21}$$

Substituting (17) and (20)-(21) in the cost functions of the generators yields for the total cost of operating the system:

$$C'_{tot} = C_1(P'_1) + C_2(P'_2) + C_3(P'_3) = 18186.67\ [£] \quad (1.22)$$

The system average cost is:

$$\frac{C'_{tot}}{P'_1+P'_2+P'_3} = 30.31\ [\frac{£}{MWh}] \quad (1.23)$$

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega'_1 = \lambda * P'_1 - C_1(P'_1) = 10560.56\ [£] \quad (1.24)$$

$$\Omega'_2 = \lambda * P'_2 - C_2(P'_2) = 1086.11\ [£] \quad (1.25)$$

$$\Omega'_3 = \lambda * P'_3 - C_3(P'_3) = -33.33\ [£] \quad (1.26)$$

[4 marks; subtract 1 mark if the calculations of optimal production are wrong; subtract 1 mark if the system marginal cost is wrong; subtract 1 mark if the system average cost is wrong; and subtract 1 mark if the calculation of generators' profits are wrong]

(d-ii) When the maximum output limits are considered, the output of generator 3 needs to be restricted to a higher value and therefore the outputs of the other two generators are reduced. Given that generator 3 is characterised by the highest variable cost, this new dispatch is more costly (i.e. it increases the total cost of operating the system, the system marginal cost and the average system cost). On the other hand, this new dispatch reduces the profit of all the generators. For generators 3 this is driven by the reduced system marginal cost and the increased generation output, while for generators 1-2 this is the reduction of both their generation outputs and the system marginal cost.

[2 marks; subtract 1 mark if the explanation of effect on the total operation cost is not clear; subtract 1 mark if the explanation of effect on the generators' profits is not clear]

(e) The negative profit of generator 3 means that the operating cost of this generator is larger than its revenue in the optimal solution and thus it experiences economic losses at the optimal solution (does not recover its operating costs). This is happening since marginal system cost pricing does not guarantee the recovery of the fixed (no-load) costs of the participating generators in combination with the high fixed costs of generator 3. This lack of recovery of some of the generators' fixed costs will mean in the long term these generators would leave the market, resulting potentially in higher total costs of operating the system or even security problems (inability of the system to satisfy the demand). Therefore, a suitable mechanism compensating the generators' unrecovered fixed costs may be required (or the generators would need modify their bids in order to recover their cost).

[3 marks; subtract 2 marks if the explanation of negative profit is not clear; subtract 1 mark if the practice to avoid the negative profit is not clear]

Answers

Solution to Question 2

(a) In a perfectly competitive market no participant has the ability to influence the market price through its individual actions (i.e. all participants act as price takers). This assumption is valid if the number of market participants is large and if none of them controls a large proportion of the production or consumption. Perfect competition is a highly desirable goal because it encourages efficient market behaviour.

[1 mark; to explain the conditions for perfect competition]

Strategic market players can manipulate the prices either by withholding quantity (physical withholding) or by raising the asking price (economic withholding), compared to the perfectly competitive case.

[1 mark; to explain how a company in an imperfect competition environment can influence the market price]

In the presence of imperfect competition market prices tend to be higher than producers' marginal costs (which determine the price in perfectly competitive markets).

[2 marks; subtract 1 mark for each condition (perfectly and imperfectly competitive markets) what is the relationship between the marginal cost of production and the market price]

(b) Under perfect competition, the price is equal to the marginal cost of each generator. Therefore, the following two equations hold:

$$\frac{\partial C_A(P_A)}{\partial P_A} = \pi = 26 + 1.0 * P_A \quad (2.1)$$

$$\frac{\partial C_B(P_B)}{\partial P_B} = \pi = 30 + 0.6 \cdot P_B \quad (2.2)$$

Since $\pi = 200 - 1.5 * D$ and $D = P_A + P_B$ (demand-supply balance condition), equations (2.1) and (2.2) yield respectively:

$$26 + 1.0 * P_A = 200 - 1.5 * (P_A + P_B) \rightarrow P_A = 69.6 - 0.6 \cdot P_B \quad (2.3)$$

$$30 + 0.3 * P_B = 200 - 1.5 * (P_A + P_B) \rightarrow P_B = 80.9524 - 0.7143 * P_A \quad (2.4)$$

(b-iii) The combination of (2.3) and (2.4) yields:

$$P_A = 36.80\ [MW] \quad (2.5)$$

$$P_B = 54.67\ [MW] \quad (2.6)$$

(b-ii) The total demand is given by:

$$D = P_A + P_B \rightarrow D = 91.47\ [MW] \quad (2.7)$$

(b-i) The electricity price is given by:

$$\pi = 200 - 1.5 \cdot D \rightarrow \pi = 62.80\ [\tfrac{£}{MWh}] \quad (2.8)$$

(b-iii) The profits made by the two generators are given by:

$$\Omega_A = P_A \cdot \pi - 26 \cdot P_A - 0.5 \cdot P_A^2 \rightarrow \Omega_A = 677.12\ [\tfrac{£}{h}] \quad (2.9)$$

$$\Omega_B = P_B \cdot \pi - 30 \cdot P_B - 0.3 \cdot P_B^2 \rightarrow \Omega_B = 896.53[\tfrac{£}{h}] \quad (2.10)$$

(b-i) [1 mark; to calculate the electricity price]

(b-ii) [1 mark; to calculate the level of demand]

(b-iii) [2 marks; subtract 1 mark if the calculation of production levels is wrong; subtract 1 mark if the calculation of profits is wrong]

(c-i) In the Cournot model of competition the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column correspond to a given production by company A and the cells in a row correspond to a given production by company B.

Each cell contains four pieces of information arranged in the following format:

D	Ω_A
Ω_B	π

where:

π price $[\frac{£}{MWh}]$

D demand $[MW]$

Ω_A profit made by company A $[\frac{£}{h}]$

Ω_B profit made by company B $[\frac{£}{h}]$

Given the productions P_A and P_B of the two companies, the other quantities are calculated as follows:

$$D = P_A + P_B \quad (2.11)$$

$$\pi = 200 - 1.5 \cdot D \quad (2.12)$$

$$\Omega_A = P_A * \pi - 26 \cdot P_A - 0.5 \cdot P_A^2 \quad (2.13)$$

$$\Omega_B = P_B * \pi - 30 \cdot P_B - 0.3 \cdot P_B^2 \quad (2.14)$$

Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below.

P_B/P_A	35		37		39	
37	**72**	**1697.5**	74	1646.5	76	1579.5
	1883.3	**92**	1772.30	89	1661.3	86
39	74	1592.5	76	2652	78	2648.6
	1844.7	89	1727.7	86	1610.7	83
41	76	1487.5	78	1424.5	80	1345.5
	1791.7	86	1668.7	83	1545.7	80

Answers

A Nash equilibrium state implies that given the production level of either of the two companies in this state, the other company cannot achieve a higher profit by following a production level different than the one at the equilibrium. It can thus be observed from the table below that the cell corresponding to $P_A = 35MW$ and $P_B = 37MW$ is an equilibrium point. In other words, given that $P_A = 35MW$, company B achieves its highest profit by producing $P_B = 37MW$ AND given that $P_B = 37MW$, company A achieves its highest profit by producing $P_A = 35MW$.

[8 marks; subtract 1 mark if the calculation steps of Nash equilibrium point are not clear; subtract 6 marks if the calculations of the table are wrong; subtract 1 mark if the Nash equilibrium point is not found]

(c-ii) We can observe that under imperfect competition the market price and the generation profits are significantly higher than under perfect competition, since producers exercise market power and manipulate the prices to strategically increase their profits. As a result of the price elasticity of the demand, the increased price leads to lower demand (and consequently to lower generation levels).

[2 marks; subtract 2 mark if the explanations of the impacts of imperfect competition on the market outcome are not clear]

(d) The key difference between Bertrand and Cournot competition models is that the first one models imperfect competition by introducing competition on prices, and the latter assumes competition takes place through companies choosing which quantity to produce in order to maximise profits.

[2 marks; subtract 2 mark if the explanations of Bertrand and Cournot competition models are not clear]

Answers

Solutions to Question 3

(a-i) Since the generators' marginal costs are fixed, the minimum cost dispatch can be calculated by dispatching the generators in an ascending marginal cost order until the total demand of $D_1 + D_2 + D_3 = 500\ [MW]$ is satisfied. We thus get for the minimum cost dispatch:

$$P_a = 155\ [MW] \tag{3.1}$$

$$P_b = 345\ [MW] \tag{3.2}$$

$$P_c = 0\ [MW] \tag{3.3}$$

$$P_d = 0\ [MW] \tag{3.4}$$

[2 marks; subtract 0.5 mark for any incorrect calculation in minimum cost dispatch of generators up to a maximum of subtracted 2 marks]

(a-ii) The line flow problem can be solved using the superposition principle and also directly. To this effect, we write the power balance equation at two buses and KVL around the loop.

$$\text{Bus 1:}\quad P_a + P_b - 80 = F_{12} + F_{13} \tag{3.5}$$

$$\text{Bus 2:}\quad P_c - 90 = -F_{12} + F_{23} \tag{3.6}$$

$$\text{Bus 3:}\quad P_d - 330 = -F_{13} - F_{23} \tag{3.7}$$

$$\text{Loop equation: } 0.3F_{12} + 0.2F_{23} - 0.3F_{13} = 0 \tag{3.8}$$

Putting these equations in the matrix form gives:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_c - 90 \\ P_d - 330 \\ 0 \end{bmatrix} \tag{3.9}$$

Substituting (3.1)-(3.4) in (3.9), we get:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} -90 \\ -330 \\ 0 \end{bmatrix} \tag{3.10}$$

Solving these equations, we get:

$$F_{12} = 180\ [MW] \tag{3.11}$$

$$F_{13} = 240\ [MW] \tag{3.12}$$

$$F_{23} = 90[MW] \tag{3.13}$$

Since $F_{12} > F_{12}^{max}$ ($F_{12}^{max} = 160\ [MW]$ according to the data) the branch 1-2 is overloaded.

[2 marks; subtract 1 mark if the calculations of power flows are wrong; subtract 1 mark if the overloaded branch is wrong]

(b-i)

Option-1) The first method consists in increasing the output of generator G_d at node 3 and decreasing by the same amount the output of generator G_a at node 1. To calculate how big this increase should be to remove the violation of the flow limit on line 1-2, consider an injection of $+1[MW]$ at node 3 and an injection of $-1[MW]$ at node 1. This pair of injection causes a flow in the network that divides itself as follows:

Answers

$\frac{0.3}{(0.3+0.2)+0.3} \cdot 1 = 0.375[MW]$ along the path 3-2-1 (3.14)

$\frac{(0.3+0.2)}{(0.3+0.2)+0.3} \cdot 1 = 0.625[MW]$ along the path 3-1 (3.15)

Since we use a linear (dc) model, we can say that to remove $F_{12} - F_{12}^{max} = 20\ [MW]$ overload on line 1-2, we therefore need to increase the output of generator G_d by:

$\frac{20}{0.375} = 53.33[MW]$ (3.16)

Thus, the new generators' dispatch becomes:

$P'_a = 101.67\ [MW]$ (3.17)

$P'_b = 345\ [MW]$ (3.18)

$P'_c = 0\ [MW]$ (3.19)

$P'_d = 53.33[MW]$ (3.20)

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F'_{12} \\ F'_{13} \\ F'_{23} \end{bmatrix} = \begin{bmatrix} P'_a + P'_b - 80 \\ P'_d - 330 \\ 0 \end{bmatrix} \quad (3.21)$$

We get:

$F'_{12} = 160\ [MW]$ (3.22)

$F'_{13} = 206.67\ [MW]$ (3.23)

$F'_{23} = 70[MW]$ (3.24)

which satisfy the lines' capacities constraints.

The cost of this dispatch is:

$C_{Total} = 101.67 * 19 + 345 * 15 + 0 * 35 + 53.33 * 25 = 8439.98\ [£]$ (3.25)

Option-2) The second method consists in increasing the output of generator G_c at node 2 and decreasing by the same amount the output of generator G_a at node 1. To calculate how big this increase should be to remove the violation of the flow limit on line 1-2, consider an injection of $+1[MW]$ at node 2 and an injection of $-1[MW]$ at node 1. This pair of injection causes a flow in the network that divides itself as follows:

$\frac{0.3}{(0.3+0.2)+0.3} \cdot 1 = 0.375[MW]$ along the path 2-3-1 (3.26)

$\frac{(0.3+0.2)}{(0.3+0.2)+0.3} \cdot 1 = 0.625[MW]$ along the path 2-1 (3.27)

Since we use a linear (dc) model, we can say that to remove $F_{12} - F_{12}^{max} = 20\ [MW]$ overload on line 1-2, we therefore need to increase the output of G_c by:

$\frac{20}{0.625} = 32[MW]$ (3.28)

Thus, the new generators' dispatch becomes:

$P'_a = 155 - 32 = 123\ [MW]$ (3.29)

$P'_b = 345\ [MW]$ (3.30)

$P'_c = 32\ [MW]$ (3.31)

$P'_d = 0[MW]$ (3.32)

Answers

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F'_{12} \\ F'_{13} \\ F'_{23} \end{bmatrix} = \begin{bmatrix} P'_a + P'_b - 80 \\ P'_c - 90 \\ 0 \end{bmatrix} \tag{3.33}$$

We get:

$$F'_{12} = 160\ [MW] \tag{3.34}$$

$$F'_{13} = 228\ [MW] \tag{3.35}$$

$$F'_{23} = 102[MW] \tag{3.36}$$

which satisfy the lines' capacities constraints.

The cost of this dispatch is:

$$C_{Total} = 123 * 19 + 345 * 15 + 32 * 35 + 0 * 25 = 8632\ [£] \tag{3.37}$$

It can be concluded that the first constrained dispatch is preferable to the second one because its cost is smaller ($8439.98 \leq 8632$). It is thus the optimal constrained dispatch.

[6 marks; subtract 3 marks if the calculations of first method are wrong; subtract 3 marks if the calculations of second method are wrong]

(b-ii) The nodal price at each bus is given by the cost of one additional MW of load at each node. Therefore, the price at node 1 is 19 $\left[\frac{£}{MWh}\right]$ because the next MW of load would be generated locally by generator G_a because it is the cheapest generator not operating at its upper limit. An additional MW of load at node 3 would have to be produced by generator G_d. The price at node 3 is therefore 25 $\left[\frac{£}{MWh}\right]$. Increasing the output of generator G_c will not lead to such a violation but seems as a costly alternative since in that case the price at node 2 is:

$$\pi_2 = MC_c = 35\ [£/\text{MWh}] \tag{3.38}$$

Our goal is to check whether a combination of modifications in the outputs of generators G_a and G_d can lead to a lower price than the one in (3.38).

Extracting an additional $1[MW]$ at node 2 and generating it at node 1 with G_a causes the following change in the flow on line 1-2:

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 0.625 \\ 0.375 \\ -0.375 \end{bmatrix} \tag{3.39}$$

Similarly, extracting an additional $1[MW]$ at node 2 and generating it at node 3 with G_d causes the following change in the flow on line 1-2:

$$\begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.25 \\ -0.75 \end{bmatrix} \tag{3.40}$$

Therefore, increase of 1 MW of the output of G_a and G_d will increase the flow in line 1-2 by 0.625 MW and 0.25 MW respectively.

We need a total increase of 1 MW in the outputs of G_a and G_d (3.28) but without increasing the power flow in line 1-2 (3.42):

$$\Delta P_a + \Delta P_d = 1\ [MW] \quad (3.41)$$

$$0.625 * \Delta P_a + 0.25 * \Delta P_d = 0 \quad (3.42)$$

which yields:

$$\Delta P_a = -0.6667[MW] \quad (3.43)$$

$$\Delta P_d = 1.6667\ [MW] \quad (3.44)$$

When these outputs' modifications are carried out the price at node 2 is:

$$\pi_2' = -0.6667 * MC_A + 1.6667 * MC_D = 29[\text{£/MWh}] \quad (3.45)$$

Which is accepted as the final value of price at node 2 since it is lower than the value in (3.38).

Finally:

$$\pi_1 = 19[\tfrac{£}{MWh}] \quad (3.46)$$

$$\pi_2 = 29[\tfrac{£}{MWh}] \quad (3.47)$$

$$\pi_3 = 25[\tfrac{£}{MWh}] \quad (3.48)$$

[4 marks; subtract 2 marks if the calculation of new electricity price at node 2 is wrong; subtract 2 marks if the new electricity prices at nodes 1&3 are wrong]

(c-i) Although the price at bus 2 is 29 [£/MWh] (3.47), the contract for difference that Load at node 2 has made at the price of 26.5 [£/MWh] means that it will pay for satisfying its demand:

$$C_2 = 29 * D_2 + (26.5 - 29) * D_2 = 2385\ [£/h] \quad (3.49)$$

Its income from the transmission congestion contract (regarding line 1-2) is given by the product of the volume of the flow in the contract (90 $[MW]$) and the price differential in the two buses where line 1-2 is connected. In other words:

$$I_2 = 90 * (\pi_2' - \pi_1) \quad (3.50)$$

Since generator G_a is the cheapest generator whose output has not reached its maximum limit (3.17)-(3.20), the price at bus 1 can be easily shown to be:

$$\pi_1 = MC_a = 19\ [\text{£/MWh}] \quad (3.51)$$

Substituting (3.47) and (3.51) in (3.50) gives:

$$I_2 = 900\ [£/h] \quad (3.52)$$

[3 marks; subtract 2 mark if the formulation of income from the transmission congestion contract is wrong; subtract 1 mark if the calculation is wrong]

(c-ii) Therefore, the net cost that load at bus 2 has to pay is (from (3.49) and (3.52)):

$$C_2^{net} = C_2 - I_2 = 1485\ [£/h] \quad (3.53)$$

and the unit price that demand at bus 2 will pay is:

$$\pi_2^{net} = \frac{C_2^{net}}{D_2} = 16.5\ [\text{£/MWh}] \quad (3.54)$$

[3 marks; subtract 2 marks if net cost that load at bus 2 has to pay is wrong; subtract 1 mark if the calculation of unit price at bus 2 is wrong]

Answers

Solution to Question 4

(a) We neglect the transmission line constraint.

Winter

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ $subject\ to$ $P_A + P_B = 3500$

$$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 3500) \quad (4.1)$$

$$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002} \quad (4.2)$$

$$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004} \quad (4.3)$$

$$P_A + P_B = 3500 \rightarrow \lambda = 6\frac{\$}{MWh} = MC_A = MC_B, P_A = 2500MW, P_B = 1000MW \quad (4.4)$$

Summer

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ $subject\ to$ $P_A + P_B = 2000$

$$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 2000) \quad (4.5)$$

$$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002} \quad (4.6)$$

$$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004} \quad (4.7)$$

$$P_A + P_B = 2000 \rightarrow \lambda = 4\frac{\$}{MWh} = MC_A = MC_B, P_A = 1500MW, P_B = 500MW \quad (4.8)$$

[4 marks; subtract 1 mark for any incorrect calculation in optimal generation in areas A and B and for winter and summer seasons up to a maximum of subtracted 4 marks]

(b) We take into account the transmission line constraint of 1200MW.

Winter

The optimal dispatch becomes: $P_A = 1200MW,\ P_B = 2300MW$ and the prices are:

$$MC_A = 3.4\frac{£}{MWh} = PriceA \quad (4.9)$$

$$MC_B = 11.2\frac{£}{MWh} = PriceB \quad (4.10)$$

$$C_A = 2{,}640\frac{£}{h},\ C_B = 15{,}180\frac{£}{h} \quad (4.11)$$

Summer

The optimal dispatch becomes: $P_A = 1200MW,\ P_B = 800MW$ and the prices are

$$MC_A = 3.4\frac{£}{MWh} = PriceA \quad (4.12)$$

$$MC_B = 5.2\frac{£}{MWh} = PriceB \quad (4.13)$$

Answers

$$C_A = 2{,}640\frac{£}{h},\ C_B = 2{,}880\frac{£}{h} \tag{4.14}$$

(b-i) [2 marks; subtract 1 mark for any incorrect calculation in electricity prices in areas A and B up to a maximum of subtracted 2 marks]

(b-ii) [2 marks; subtract 1 mark for any incorrect calculation in generation cost in areas A and B up to a maximum of subtracted 2 marks]

(c) (i) We take into account the transmission line constraint of 1200MW.

The annual generation costs are:

$$C_A^{winter} = 2{,}640\frac{£}{h} \cdot 2700h = 7{,}128{,}000\ £ \tag{4.15}$$

$$C_B^{winter} = 15{,}180\frac{£}{h} \cdot 2700h = 40{,}986{,}000\ £ \tag{4.16}$$

$$C_A^{summer} = 2{,}640\frac{£}{h} \cdot 6060h = 15{,}998{,}400\ £ \tag{4.17}$$

$$C_B^{summer} = 2{,}880\frac{£}{h} \cdot 6060h = 17{,}452{,}800\ £ \tag{4.18}$$

$$C_{A+B}^{annual} = 81{,}565{,}200\ £ \tag{4.19}$$

[2 marks; subtract 1 mark if any calculation of annual generation cost in areas A and B for two seasons is wrong; subtract 1 mark if the total annual generation costs are wrong]

(c) (ii) The electricity price in winter is 11.2 £/MWh and in summer is 5.2 £/MWh.

The annual demand charges are:

$$3500MW \cdot 2700h \cdot 11.2\frac{£}{MWh} + 2000MW \cdot 6060h \cdot 5.2\frac{£}{MWh} = 168{,}864{,}000£ \tag{4.20}$$

The annual generation revenues are:

$$R_A^{winter} = 1200MW \cdot 3.4\frac{£}{MWh} \cdot 2700h = 11{,}016{,}000\ £ \tag{4.21}$$

$$R_B^{winter} = 2{,}300MW \cdot 11.2\frac{£}{MWh} \cdot 2700h = 69{,}552{,}000\ £ \tag{4.22}$$

$$R_A^{summer} = 1200MW \cdot 3.4\frac{£}{MWh} \cdot 6060h = 24{,}724{,}800\ £ \tag{4.23}$$

$$R_B^{summer} = 800MW \cdot 5.2\frac{£}{MWh} \cdot 6060h = 25{,}209{,}600\ £ \tag{4.24}$$

$$R_{A+B}^{annual} = 130{,}502{,}400\ £ \tag{4.25}$$

[2 marks; subtract 1 mark if the annual demand charges are wrong; subtract 1 mark if the total annual generation revenues are wrong]

(c) (iii) The TransCo's total annual revenue is

$$R_{Tra} = 1200MW(11.2 - 3.4) \cdot 2700h + 1200 \cdot (5.2 - 3.4) \cdot 6060 = 38{,}361{,}600£ \tag{4.26}$$

We can see that it is equal to the difference between annual demand charges and annual generation revenues.

[2 marks; to calculate the total annual transmission revenue received by Borduria Transco]

Answers

(d) After the reinforcement, the transmission line constraint becomes (1200+900) = 2100MW

Winter

The optimal dispatch becomes: $P_A = 2100MW,\ P_B = 1400MW$ and the prices are:

$$MC_A = 5.2\frac{£}{MWh} = PriceA \quad (4.27)$$

$$MC_B = 7.6\frac{£}{MWh} = PriceB \quad (4.28)$$

Summer

The optimal dispatch becomes: $P_A = 1500MW,\ P_B = 500MW$ and the prices are

$$MC_A = 4\frac{£}{MWh} = PriceA \quad (4.29)$$

$$MC_B = 4\frac{£}{MWh} = PriceB \quad (4.30)$$

Thus, Borduria TransCo's total annual revenue will be:

$$R_{Tra} = 2100MW \cdot (7.6 - 5.2) \cdot 2700h = 13{,}608{,}000\ £ \quad (4.31)$$

The reinforcement will cost £5,000,000.

Also, total annual demand charges:

$$3500 \cdot 2700 \cdot 7.6 + 2000 \cdot 6060 \cdot 4 = 120{,}300{,}000\ £ \quad (4.28)$$

i.e. a reduction of £48,564,000.

In addition, the total annual generation costs will be

$$3750 \cdot 6060 + 1500 \cdot 6060 + 6510 \cdot 2700 + 6720 \cdot 2700 = 67{,}536{,}000\ £ \quad (4.29)$$

i.e. a reduction of £14,029,200

This investment will benefit all parties in the system: The total generation cost will reduce (i.e. constraints cost will reduce), and total annual demand charges will also reduce.

However, still the total annual revenue of TransCo is higher than its total annuitized investment cost, signifying a degree of underinvestment, Therefore Borduria Regulator will prefer to reject this investment and instead suggest a greater investment such that neither underinvestment nor overinvestment occurs in the system.

[6 marks; subtract 2 marks if the optimal dispatches and marginal prices in areas A and B and two seasons are wrong; subtract 1 mark if Borduria TransCo's total annual revenue is wrong; subtract 1 mark if total annual demand charge is wrong; subtract 1 mark if total annual generation cost is wrong; and subtract 1 mark if explanation of the proposed investment strategy is not clear]

Answers